

De Novo Design of Miniprotein Candidate Neutralizers Targeting the IL-23 Cytokine Subunit p19: A Rigorously Scored Pipeline Comparison

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Abstract—Interleukin-23 (IL-23) is a validated driver of T-helper-17-mediated inflammation in inflammatory bowel disease and psoriasis, and is neutralized clinically by anti-p19 antibodies. De novo miniproteins offer a smaller, more stable, potentially oral modality, but reported designs target the IL-23 *receptor* (IL-23R); the IL-23 *cytokine* itself has not been addressed by de novo design. We computationally design miniprotein binders against the IL-23R-binding face of the p19 subunit, using PDB 3DUH as the target. Scoring every design with an interface-aware metric (AlphaFold-initial-guess predicted aligned error processed by ipSAE), we find that a conventional RFdiffusion → ProteinMPNN → ColabFold pipeline produces no reliable binder across 972 designs (best ipSAE 0.30; none above the 0.60 reliable-interface threshold), and we show that the pessimistic single-sequence ColabFold signal that initially suggested unfoldable binders is a scoring artifact, where there was a ~ 56 -point binder-pLDDT under-rating relative to AlphaFold-initial-guess on identical backbones. An interface-optimized hallucination method (BindCraft), scored by the identical pipeline, yields 42 of 50 designs above the 0.60 reliable-interface threshold (38 in the ≥ 0.70 confident band; best ipSAE 0.804). We characterize five all- α -helical leads (72–104 residues) that together engage all six targeted epitope hotspots; two dock within the intact p40/p19 heterodimer, and all show favorable Rosetta interface energetics (ΔG -57 to -87 REU). To our knowledge these are the first de novo miniprotein candidates aimed at neutralizing the IL-23 cytokine directly. All results are computational; experimental validation is future work.

Index Terms—de novo protein design, protein-protein interface, AlphaFold, ipSAE, IL-23, inflammatory bowel disease, miniprotein therapeutics.

I. INTRODUCTION

Interleukin-23 (IL-23) is a heterodimeric pro-inflammatory cytokine composed of a shared p40 subunit and an IL-23-specific p19 subunit, the latter a four-helix-bundle cytokine. IL-23 sustains pathogenic T-helper-17 responses and is among the most thoroughly validated therapeutic targets in inflammatory bowel disease (IBD) and psoriasis. Monoclonal antibodies that bind p19 and neutralize the cytokine, such as risankizumab and guselkumab among them, are approved and effective [14], but share the limitations of antibody biologics: high manufacturing cost, parenteral administration, and cold-chain logistics.

De novo–designed miniproteins are an attractive alternative modality. At roughly a tenth the size of an antibody, hyperstable designed binders can resist heat, acid, and proteolysis [16], opening the door to low-cost manufacturing and even oral delivery. Recent work demonstrated this concretely for the IL-23 axis: designed miniproteins that inhibit the IL-23

receptor (IL-23R) and the IL-17 cytokine reached antibody-like potency, with an orally delivered IL-23R minibinder outperforming a clinical anti-IL-23 antibody in mouse colitis [9]. That work targets the receptor and IL-17A. The complementary target (the IL-23 *cytokine* itself, neutralized at its IL-23R-binding face in the manner of an anti-p19 antibody) has not, to our knowledge, been addressed by de novo miniprotein design. That is the gap this paper addresses.

Designing a binder is only half the problem; deciding whether a design is likely to work is the other half, and is where in-silico campaigns most often go wrong. AlphaFold-Multimer’s interface confidence score (ipTM) is widely used but is computed over whole chains, so it inflates or deflates with construct length rather than tracking the interface itself [2]. The recently introduced ipSAE metric corrects this by restricting the template-modeling expression to confidently predicted interchain residue pairs, separating true from false complexes more reliably than ipTM [8]. We adopt ipSAE, interpreting it against established confidence bands [15] (with a reliable-interface threshold of 0.60 and the ≥ 0.70 confident band reported alongside) as our primary success criterion, and apply the *same* scorer to every design from every pipeline so that comparisons are fair.

This paper makes three contributions. First, we characterize a scoring artifact that can derail binder-design campaigns: single-sequence ColabFold systematically and severely under-rates designed binders relative to AlphaFold-initial-guess, producing a spurious “binders don’t fold” signal. Second, using a rigorous, interface-aware scorer, we show that a standard RFdiffusion/ProteinMPNN/ColabFold pipeline fails to produce a single reliable binder against the IL-23R face of p19 across 972 designs, an evident negative result. Third, we show that switching to interface-optimized hallucination (BindCraft), with identical scoring, yields 42 of 50 designs above the 0.60 reliability threshold (38 in the ≥ 0.70 confident band), and we structurally and energetically characterize five leads that collectively cover the targeted epitope, including two that engage p19 within the intact heterodimer.

II. RELATED WORK

Deep-learning protein design. Backbone generation by RFdiffusion [4], sequence design by ProteinMPNN [5], and structure prediction by AlphaFold2 [1] and AlphaFold-Multimer [2] (with the ColabFold [3] implementation widely used for accessibility) together form the dominant de

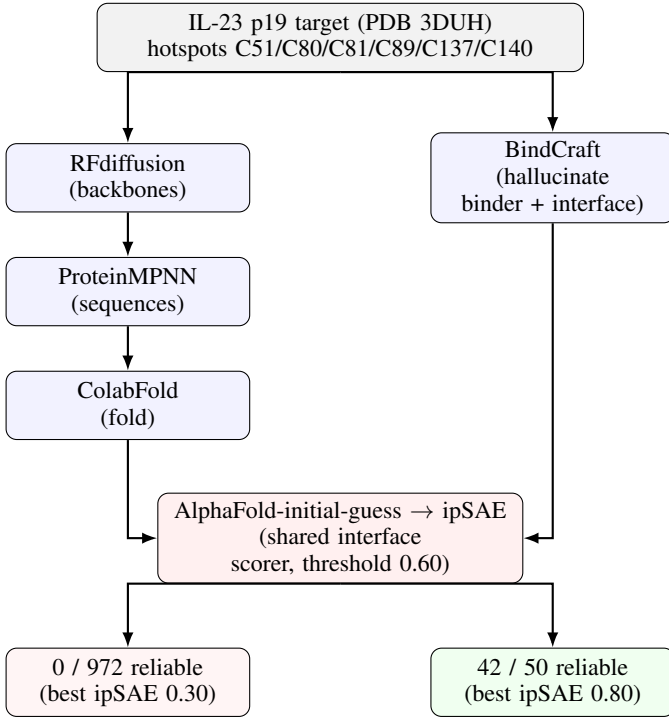


Fig. 1: Pipeline overview. Two structure-conditioned design tracks (a conventional RFdiffusion/ProteinMPNN/ColabFold pipeline and interface-optimized BindCraft hallucination) feed an *identical* AlphaFold-initial-guess + ipSAE interface scorer (reliable-interface threshold 0.60). Under the shared scorer the two tracks give opposite outcomes.

novo binder-design stack. AlphaFold-initial-guess adapts AlphaFold2 to evaluate designed complexes by seeding the prediction with the designed pose, and was shown to raise experimental binder-design success rates roughly tenfold when used as a filter [6]. BindCraft takes a different route, hallucinating binder and interface together by backpropagation through AlphaFold2 and co-folding target and binder to allow induced fit, reporting experimental success rates of 10–100% across diverse targets [7].

Interface confidence metrics. Beyond ipTM, two recent metrics target its length sensitivity: actipfTM restricts the pTM calculation to flexible, interface-participating residues [13], and ipSAE derives a template-modeling-style score from the predicted aligned error over confidently predicted interchain pairs [8]. We use ipSAE as our primary metric and reproduce the reference implementation exactly on its published example.

IL-23 biology and miniprotein therapeutics. The 2.3-Å crystal structure of human IL-23 (PDB 3DUH) established the p19/p40 architecture and the p19 four-helix bundle [10]. Designed miniproteins against IL-23R and IL-17A established the feasibility of the miniprotein modality for the Th17 axis [9]; our work extends that modality to the IL-23 cytokine.

III. METHODS

A. Target preparation. IL-23 coordinates were taken from PDB 3DUH (chain A = p40, chain C = p19; p19 modeled at residues 1–29, 47–94, 100–125, 137–169). We defined two targets: the isolated p19 cytokine (chain C, 137 modeled residues) and the intact p40/p19 heterodimer. Six residues on the IL-23R-binding face of p19 were selected as design hotspots: C51, C80, C81, C89, C137, C140 (3DUH chain-C numbering); all lie in modeled regions.

B. Conventional pipeline (baseline). Binder backbones were generated with RFdiffusion conditioned on the p19 hotspots (1,000 backbones; 972 valid after filtering and sequence threading), sequences were assigned with ProteinMPNN, and complexes were initially folded with single-sequence ColabFold.

C. Interface-aware re-scoring. Every design was re-evaluated with AlphaFold-initial-guess, seeding the binder–p19 complex and recovering per-residue pLDDT and the full predicted-aligned-error (PAE) matrix. From the PAE we computed ipSAE following the reference definition: for each residue, interchain partners with PAE below the cutoff are counted, a distance scale d_0 is derived from that count (with a residue-count floor), and a template-modeling-style transform is averaged over valid pairs; the pair score is the maximum over residues and over both chain directions. We report ipSAE as the maximum over the two asymmetric chain directions, using Dunbrack’s Eq. 16 and the headline value emitted by the reference `ipsae.py` [8]. Our implementation reproduces the `ipsae.py` output to full numerical agreement on its distributed example, used solely as a code-correctness check, not a threshold rationale. For interpretation we adopt the AlphaFold-DB ipSAE confidence bands: ≥ 0.80 very high; 0.70–0.80 confident; 0.60–0.70 possibly correct but to be read cautiously; < 0.60 unlikely to be a reliable interface [15] (bands defined for homodimer interfaces; applied here to the binder–p19 heterodimer). Thus, we take $\text{ipSAE} \geq 0.60$ as the reliable-interface threshold and report the ≥ 0.70 confident-band count alongside. Because the maximum is by construction at least the per-direction minimum, applying the 0.60 threshold to the maximum is permissive relative to a minimum-based operating point; this affects neither the five leads (all ≥ 0.70 on the reported value) nor the conventional-versus-BindCraft contrast, since the conventional best (0.297) falls below every threshold under either convention.

D. Interface-optimized pipeline (BindCraft). BindCraft was run against both the p19-only and the full-heterodimer targets with the same six hotspots, using the `default_4stage_multimer_flexible_hardtarget` protocol, which is BindCraft’s four-stage routine in which the binder is hallucinated against the co-folded target by backpropagation through AlphaFold2-multimer, its sequence is redesigned with ProteinMPNN, and the complex is re-predicted and filtered, run with a flexibly modeled (relaxable) target and the stricter “hard-target” acceptance criteria; the standard protocol collapsed at the

sequence-discretization stage on this target, which is why the flexible-hardtarget variant was required, and a configured binder-length range of 65–130 residues. BindCraft reports per-design Rosetta interface metrics (ΔG , buried surface area, shape complementarity, packing, interface hydrogen bonds, hotspot RMSD, helicity).

E. Apples-to-apples comparison. Every BindCraft *Accepted* design was re-scored through the *identical* AlphaFold-initial-guess + ipSAE pipeline used for the 972 conventional designs (binder relabeled, p19 = the C-terminal 137 residues of the target). Only the design method differs between the two campaigns; the scorer is held fixed.

F. Lead characterization. For each lead we determined interface contacts (binder heavy atom within 5 Å of p19), mapped them to canonical 3DUH chain-C numbering to identify engaged hotspots, computed secondary structure by DSSP (via PyRosetta [11], [12]), and recorded the BindCraft Rosetta interface metrics.

G. Compute. All design and scoring ran on an HPC cluster with NVIDIA A100 (40 GB) GPUs under SLURM.

IV. RESULTS

A. The ColabFold scoring artifact

An early reading of the conventional pipeline suggested the designed binders were intrinsically disordered: single-sequence ColabFold assigned them very low binder pLDDT. Re-folding the *same* backbones with AlphaFold-initial-guess overturned this. On 50 matched designs, mean binder pLDDT was 30.1 under single-sequence ColabFold versus 86.3 under AlphaFold-initial-guess, which is a mean under-rating of 56 points (per-design range +30 to +79; Fig. 2). ColabFold’s ipTM was, moreover, essentially uncorrelated with ipSAE on these designs (Spearman ≈ -0.07). The “unfoldable binder” signal was therefore an artifact of single-sequence scoring, not evidence of real instability; the binders fold confidently, and the open question is the *interface*, not the monomer.

B. The conventional pipeline finds no reliable binder

With foldability resolved, we scored interfaces directly. Across the full 972-design screen, no design reached the reliable-interface threshold: the best binder-p19 ipSAE was 0.297, the median was 0.005, and 0/972 reached 0.60 (none reached even 0.40), with the entire distribution collapsed toward zero (Fig. 3). A curated top-50 subset, re-scored against both IL-23 subunits, confirmed that the designs engaged neither: the best p19 interface (ipSAE 0.235) and the best off-target p40 interface (0.147) both fell well below threshold. Because the same designs fold well (AlphaFold-initial-guess binder pLDDT ~ 82 –86) yet form no reliable interface, the limitation is interface optimization, not backbone quality, which is precisely the failure mode an interface-aware scorer is meant to expose.

C. Interface-optimized design clears the reliability threshold

Re-running the campaign with BindCraft, and scoring with the identical AlphaFold-initial-guess + ipSAE pipeline,

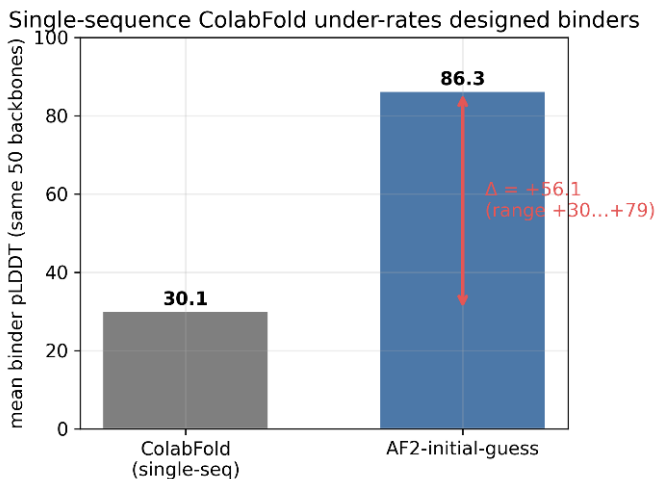


Fig. 2: The scoring artifact. Mean binder pLDDT over 50 identical backbones: single-sequence ColabFold (30.1) versus AlphaFold-initial-guess (86.3), a 56-point gap (per-design range +30 to +79).

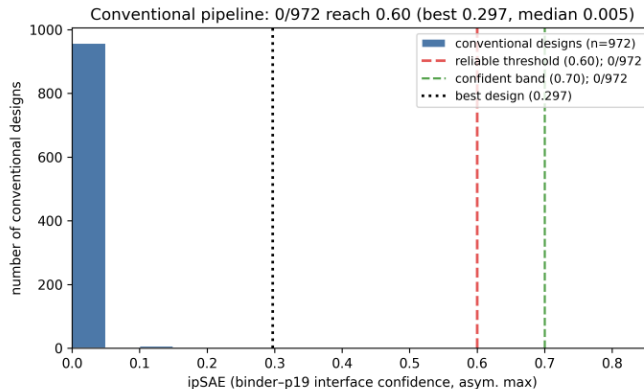


Fig. 3: Conventional pipeline ipSAE distribution. All 972 RFdiffusion/ ProteinMPNN designs, scored by the identical AlphaFold-initial-guess + ipSAE pipeline. The mass piles near zero (median 0.005); the single best design reaches 0.297 (dotted), and none approach the 0.60 reliable-interface threshold (red dashed) or the 0.70 confident band (green dashed). Contrast Fig. 4a for the BindCraft designs on the identical scale.

changed the outcome sharply. Of 50 Accepted designs (42 against p19-only, 8 against the full heterodimer), 42 cleared the 0.60 reliable-interface threshold, 38 of them in the ≥ 0.70 confident band and 2 in the ≥ 0.80 very-high band, while the best reached ipSAE was 0.804, and the minimum interface PAE fell below 1 Å (Fig. 4a). The reliable designs span at least seven independent design seeds, so the result is not a single fortunate scaffold. The eight designs below 0.60 comprise the four full-heterodimer designs with the weakest interfaces (ipSAE 0.08–0.33) together with four p19-only designs (0.47–0.55); the full-heterodimer context accounts for the lowest-scoring failures.

The contrast under identical scoring is the headline result: 0/972 (best 0.297) for the conventional pipeline versus 42/50 at the 0.60 threshold (38/50 in the ≥ 0.70 confident band; best 0.804) for BindCraft.

The design landscape (Fig. 4b) shows reliable designs across the observed 66–128-residue range, with the p19-only regime almost entirely above threshold (38/42) and the four full-heterodimer failures forming the lowest-scoring tail.

Scaffold and sequence diversity. The 42 reliable designs are not a few near-duplicate solutions. They derive from 23 distinct BindCraft backbones (design seeds) spanning 19 different binder lengths between 66 and 128 residues. Across the 861 design pairs the median pairwise sequence identity is 27% (mean 28%), and only 2.2% of pairs exceed 50% identity; single-linkage clustering at 50% identity yields 23 clusters, exactly the number of distinct backbones, so the only highly similar pairs are alternative ProteinMPNN sequences of the same scaffold. The reliable set therefore represents a diverse panel of independent solutions rather than repeated sampling of one motif.

D. Five lead candidates cover the targeted epitope

We selected five leads spanning both target regimes and the full ipSAE range, and characterized each in detail (Table I). All five are compact all- α -helical bundles of 72–104 residues. No single lead engages all six hotspots, but the set collectively covers all of them (Fig. 5a): the central 80/81/89 patch is engaged by the highest-confidence p19-only leads, the C-terminal 137/140 patch by lead l104, and the full-context lead l92 spans five of six hotspots (51/80/81/89/137). Per-hotspot lead coverage is C51 $\times$ 4, C80 $\times$ 4, C81 $\times$ 4, C89 $\times$ 4, C137 $\times$ 2, C140 $\times$ 1. This indicates the entire IL-23R face is designable, not merely one sub-patch.

ipSAE (AlphaFold interface confidence) and Rosetta ΔG (energetic estimate) are independent lines of evidence, and they agree on overall quality while ranking the leads differently (Fig. 5b, Fig. 6). Lead l94 ranks first by ipSAE (0.804) yet has the smallest interface (buried area 2204 Å²) and fewest interface hydrogen bonds (3.5); lead l72 ranks third by ipSAE (0.785) but is strongest by ΔG (−87.3 REU) with the largest interface. Reporting both metrics is therefore informative: they concur that all five are good and disagree on which is best.

The most physiologically credible lead is l92, which docks on p19 within the intact p40/p19 heterodimer and engages five of six hotspots. We note one caveat: BindCraft’s own hotspot RMSD for l92 (3.17 Å) is higher than for the p19-only leads (~1.4–1.7 Å), indicating that its docking geometry deviates somewhat from the exact requested hotspot set even though it satisfies the 5-Å contact criterion at five of six positions.

E. Structural characterization

All five leads are well-formed α -helical bundles (78–91% helix, 0% β -strand by DSSP), ranging from a compact two-helix hairpin (l72, 72 residues) to three- and four/five-helix bundles. These topologies and sizes are favorable for recombinant expression and chemical synthesis. Figure 7 presents

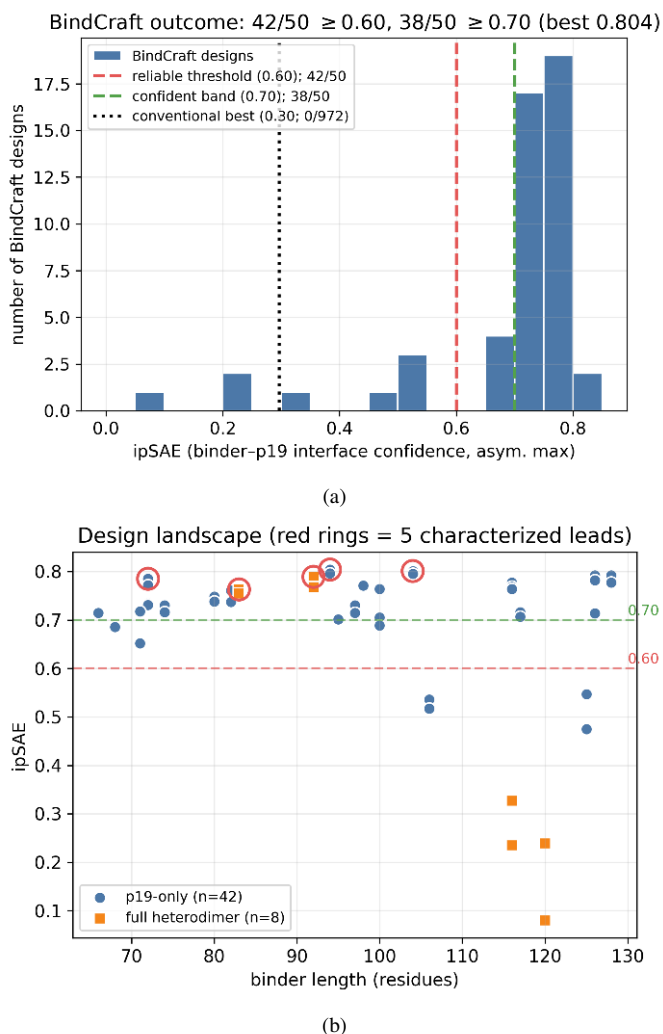


Fig. 4: BindCraft outcome under the shared scorer. (a) Distribution of binder-p19 ipSAE over 50 BindCraft designs; 42 reach the reliable-interface threshold (red dashed, 0.60) and 38 fall in the ≥ 0.70 confident band (green dashed); best 0.804. The conventional pipeline’s best (dotted, 0.297; 0/972) sits well below threshold. (b) ipSAE versus binder length for the 50 designs (p19-only versus full heterodimer); rings mark the five characterized leads.

the predicted complexes; each binder (blue) is shown on p19 (gray) with the interface residues (within 5 Å of the binder) as sticks, and p40 (tan) is shown for the two full-heterodimer leads.

F. Predicted selectivity for IL-23 over IL-12 and p40

Because IL-23 shares its p40 subunit with IL-12, specificity is a reasonable concern for any IL-23 therapeutic. Two features of the design make off-target engagement unlikely. First, the entire epitope is on p19, which is unique to IL-23, whereas p40 is the subunit shared with IL-12 [17]; a binder directed at the p19 IL-23R-binding face therefore has no cognate surface on IL-12, so cross-reactivity with IL-12 is structurally pre-

TABLE I: Five lead miniprotein candidates. ΔG in Rosetta Energy Units (REU); buried surface area dSASA in $\text{\AA}^2$; SC = shape complementarity; HB = interface hydrogen bonds; Cont. = binder-p19 heavy-atom contacts within 5 $\text{\AA}$. %Hel is DSSP-derived (Methods F); ΔG , dSASA, SC, and HB are BindCraft per-design Rosetta metrics.[†]

#	Design	Regime	Len	ipSAE	minPAE ($\text{\AA}$)	Hotspots (n/6)	%Hel	ΔG (REU)	dSASA	SC	HB	Cont.
1	I94_s482991_mpnn8	p19-only	94	0.804	0.94	51,80,81,89 (4)	91	-57.5	2204	0.68	3.5	35
2	I104_s309115_mpnn6	p19-only	104	0.801	0.99	51,137,140 (3)	91	-84.9	2962	0.69	10.5	34
3	I72_s467308_mpnn8	p19-only	72	0.785	0.92	80,81,89 (3)	88	-87.3	3018	0.68	8.5	39
4	I92_s675453_mpnn18	full	92	0.789	1.05	51,80,81,89,137 (5)	78	-77.0	2960	0.62	9.5	28
5	I83_s957191_mpnn2	full	83	0.763	1.12	51,80,81,89 (4)	81	-75.0	2943	0.67	9.0	22

[†] Designs are the five leads of § IV-D; full sequences are provided in Supplementary Data S1. ΔG , dSASA, SC, and HB are BindCraft per-design Rosetta metrics: the two full-heterodimer leads (I92, I83) from the full-heterodimer design statistics and the three p19-only leads (I94, I104, I72) from the p19-only design statistics.

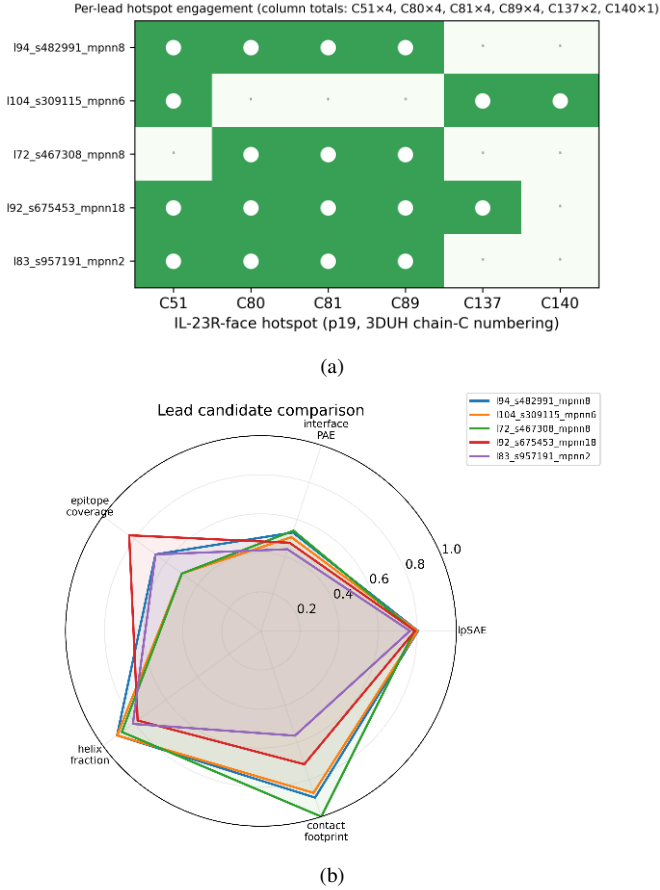


Fig. 5: Lead characterization. (a) Epitope coverage: filled circles mark engaged hotspots (binder heavy atom within 5 $\text{\AA}$ of p19, mapped to 3DUH chain-C numbering); columns total to all-six coverage across the set. (b) Multi-metric comparison of the five leads across ipSAE, interface PAE, epitope coverage, helix fraction, and contact footprint (each normalized).

cluded rather than merely improbable. This mirrors the design rationale of the clinical p19-specific antibodies, which spare p40/IL-12 unlike the p40-directed ustekinumab [17]. Second, all six design hotspots (C51/C80/C81/C89/C137/C140) lie on the p19 receptor face, away from p40: when the conventional designs were re-scored against p40, the best p40 interface

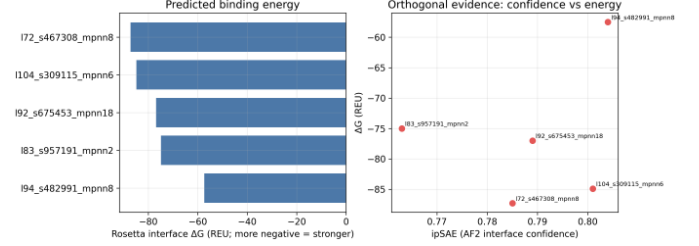


Fig. 6: Biophysics. Left: Rosetta interface ΔG per lead. Right: ipSAE versus ΔG , showing the two metrics are complementary.

reached only ipSAE 0.147 (versus a best p19 of 0.235), far below the 0.60 threshold, and the two leads optimized within the intact heterodimer localized entirely to p19 hotspots despite p40 being present. We did not, however, re-score each BindCraft lead against p40 alone, as the three p19-only leads were modeled without p40, so an explicit AlphaFold-initial-guess counter-screen of every lead against p40 and against intact IL-12 is the direct confirmatory specificity experiment and remains future work.

V. DISCUSSION

The central finding is methodological as much as biological. Two structure-conditioned design pipelines, evaluated by an identical interface-aware scorer, gave opposite outcomes on the same target: the conventional fixed-backbone pipeline produced confident folds but no reliable interface (0/972), whereas interface-optimized hallucination produced reliable interfaces at high rate (42/50; 38/50 in the ≥ 0.70 confident band). The difference is that BindCraft co-optimizes the binder and the interface and permits induced fit, while the conventional pipeline commits to a backbone before the interface is evaluated. On a flat, mostly helical cytokine face like p19's IL-23R epitope, that distinction appears decisive.

The scoring artifact (Section IV-A) is a cautionary point of independent interest. Had we trusted single-sequence ColabFold, we would have wrongly concluded the binders were unfoldable and possibly abandoned otherwise-recoverable backbones. Interface-aware, initial-guess-seeded scoring is essential not only for ranking but for avoiding false nega-

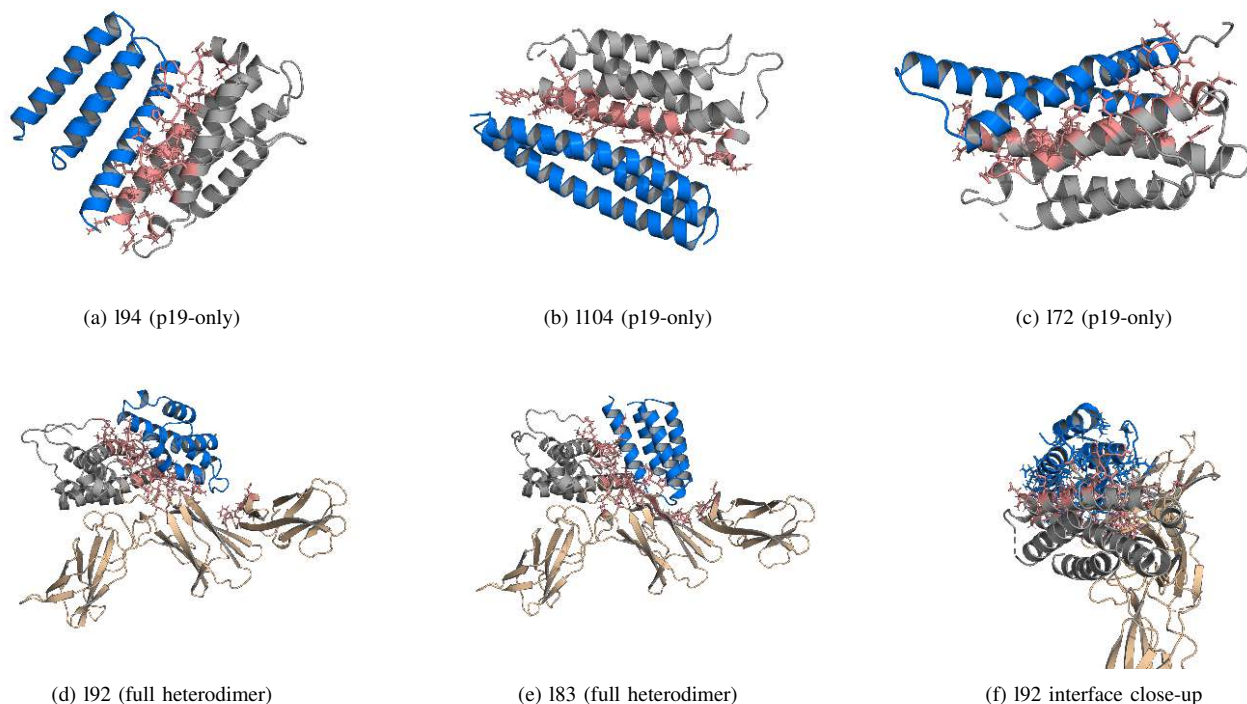


Fig. 7: Predicted lead complexes. Five lead miniproteins (blue) bound to p19 (gray), interface residues shown as sticks; for the two full-heterodimer leads p40 is tan. Flagship 192 is shown within the intact heterodimer (d), with a zoomed interface close-up (f).

tives, and ipSAE’s length-robustness matters when comparing binders of different sizes (66–128 residues here).

Biologically, the five leads collectively cover the entire targeted IL-23R face, and the complementarity of ipSAE and Rosetta ΔG gives two independent reasons for confidence while appropriately disagreeing on the single best candidate. The full-heterodimer lead 192 is the most physiologically relevant, since it engages p19 in the presence of p40, though its higher hotspot RMSD warrants scrutiny.

It is informative to place the designed epitope alongside the clinical anti-p19 antibodies. A structural survey of the p19 inhibitors found that, of the 24 p19 residues forming the IL-23R epitope, guselkumab overlaps 58% (14 residues) and risankizumab only 8% (2 residues), with the remainder of their footprints lying off the receptor site [17]. Our binders were explicitly conditioned on the IL-23R-binding face, so they most closely recapitulate the directly receptor-competitive mode exemplified by guselkumab, rather than the largely steric occlusion used by risankizumab. This both supports the mechanistic plausibility of neutralization and suggests a design knob: the six-hotspot epitope could be widened or shifted to tune between direct competition and steric blockade. A precise residue-level overlap with each antibody footprint, in a common numbering frame, would sharpen this comparison and is a natural next analysis.

Limitations. All results are computational. The evidence is AlphaFold-initial-guess interface confidence (ipSAE), Al-

phaFold pLDDT, and Rosetta energetics; ipSAE is a predictor, not a guarantee, and designed binders can still fail experimentally. The target is a single, partially modeled crystal conformation (p19 has unmodeled loops at 30–46, 95–99, 126–136). “Neutralizer” is a mechanistic hypothesis grounded in engaging the IL-23R-binding face (predicted competition with IL-23R), not a demonstrated functional block. We did not measure affinity, and BindCraft’s reported experimental success rates pertain to other targets, not ours.

Future work. The natural next steps are recombinant expression of the leads, affinity measurement by biolayer interferometry or surface plasmon resonance, a cell-based IL-23 signaling (STAT3) blockade assay to test neutralization, and ideally an experimental co-structure to confirm the predicted epitope.

VI. CONCLUSION

We presented a rigorously and uniformly scored comparison of de novo binder-design pipelines against the IL-23R-binding face of the IL-23 cytokine subunit p19. A conventional pipeline produced no reliable binder across 972 designs under interface-aware scoring, after we first showed that the discouraging single-sequence ColabFold signal was a scoring artifact. An interface-optimized pipeline, scored identically, produced 42 reliable designs out of 50 (38 in the ≥ 0.70 confident band), from which we characterized five all- α leads that collectively engage all six targeted hotspots, two within

the intact heterodimer, all with favorable interface energetics. These appear to be the first de novo miniprotein candidates aimed at neutralizing the IL-23 cytokine directly, complementing existing receptor-targeting designs and providing a starting point for experimental validation.

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